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Detection and Classification of Diabetic Retinopathy using a Federated Learning System ensuring Data Privacy

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***Abstract*: Diabetic Retinopathy (DR) continues to be a major cause of visual impairment worldwide, and there is a need for early and precise diagnosis through retinal image classification. The conventional machine learning models tend to demand centralized data storage, compromising patient privacy and data security. To overcome these issues, this study suggests a longevity focused federated learning paradigm combined with Convolutional Neural Networks for DR detection and classification. By using federated learning, our method enables several healthcare centers to jointly train a global model without the need to share confidential patient data, thereby maintaining data privacy and meeting medical data regulations. The CNN-based architecture is designed to extract deep visual features from retinal images, facilitating high-accuracy classification and accounting for model longevity using methods such as continuous learning, model adaptation, and strong aggregation techniques. We use a federated averaging algorithm that balances local updates between participating nodes to provide model stability and performance over long durations, even in changing healthcare settings with continually advancing imaging technologies. Experimental outcomes confirm that the new federated CNN model supports accuracy levels similar to, and in certain cases higher than, conventional centralized models, while largely improving data protection and model resilience. Our longevity aware design guarantees the model to be robust to shifts in data distribution and device involvement, and therefore a good fit for privacy-preserving and scalable DR diagnosis. This work sets avenues for future sustainable deployment of federated learning systems in healthcare imaging applications with geographically distributed medical centers and encourages collaborative innovation.**

Keywords: Diabetic Retinopathy, Federated Learning, Retina, Deep Learning.

I. INTRODUCTION

Diabetes Mellitus is a chronic group of metabolic disorders characterized by prolonged hyperglycemia resulting from impaired insulin secretion, action, or both. The condition is associated with various systemic complications, including cardiovascular disease, nephropathy, neuropathy, and retinopathy. Among these, Diabetic Retinopathy is one of the most common and vision-threatening microvascular complications, particularly affecting the working-age population worldwide [1].High blood sugar levels over an extended period damage the retinal microvasculature, leading to structural alterations such as microaneurysms, hemorrhages, hard exudates, and neovascularization. These changes can be observed through fundus photography, a non-invasive imaging modality frequently used in ophthalmology for DR screening [2].DR is typically classified into two main types: Non Proliferative Diabetic Retinopathy and Proliferative Diabetic Retinopathy. NPDR progresses through three stages: mild, moderate, and severe depending on the severity and extent of vascular abnormalities. PDR represents the most advanced form, characterized by neovascularization and a high risk of retinal detachment and vision loss [3].The incidence of diabetes is rising dramatically; according to the International Diabetes Federation (IDF), global diabetes prevalence is projected to increase by 46% by the year 2045. Consequently, early and automated detection of DR has become imperative to prevent blindness and reduce the global burden of diabetic complications.Historically, ophthalmologists performed manual diagnosis of retinal diseases, which was time-consuming and resource-intensive. With the advent of Artificial Intelligence (AI), particularly Machine Learning (ML) and Deep Learning (DL), automated DR detection systems have emerged, showing significant promise in identifying complex retinal patterns with high accuracy and efficiency. ML models such as Support Vector Machines (SVMs) and k-Nearest Neighbors (k-NN) rely on manual feature extraction, whereas DL techniques—especially Convolutional Neural Networks automatically learn features from data, enhancing classification performance[4].These systems simulate real-world triage by referring low-confidence predictions to specialists while automating high-confidence ones, effectively integrating into clinical workflows.Despite their advantages, calibrating prediction certainty and reducing false positives remain active research challenges.Furthermore, global DR screening programs aim to detect DR in asymptomatic individuals to initiate early treatment. However, logistical challenges and inter-patient variability complicate individual diagnosis, necessitating robust and scalable diagnostic tools [5].

II. LITERATURE SURVEY

A. Deep Learning Architectures for DR Detection :

N. Tsiknakis et al. [6],proposed a multitask deep learning framework consisting of separate classification and regression models. These models, trained independently, generate feature outputs that are concatenated and passed through a multilayer regression model, with a final perceptron layer for five-stage DR classification. S. Suganyadevi et al. [7] developed a symmetric convolutional neural network that increases both depth and width of the architecture. This design compensates for class imbalance and enhances the model's ability to detect and categorize DR severity levels without overfitting.M. Mateen et al. [8] implemented an ensemble approach by integrating multiple CNN classifiers. Their model improved accuracy and robustness against noisy inputs, demonstrating the effectiveness of combined architectures.

B. Ensemble and Hybrid Systems :

J. Wang et al. [9] introduced a meta-ensemble learning approach that combines different machine learning models for stratifying DR risk levels. The model supports clinical decision-making by outputting temporal risk scores based on patient data.K Shankar et al. [10] employed a set of nine Bayesian Neural Networks (BNNs) to estimate uncertainty in both binary and multi-class DR classifications. Their model selectively refers to uncertain predictions for human review, enhancing diagnostic reliability.

M. Shelar et al. [11] explored a combination of supervised and unsupervised learning techniques and achieved notable performance across various datasets.

III. SYSTEM ARCHITECTURE

A. Lesion Detection and Localization :

M. Z. Atwany et al. [12] used a modified Region-based Fully Convolutional Network (R-FCN) for both DR grading and lesion detection. The architecture was adapted from multi-object detection to a single-object focus, optimizing for DR grading.

W. L. Alyoubi et al. [13] extended R-FCN for lesion detection and grading, adjusting the architecture to include a softmax classification layer for single-output tasks.S. Stolte et al. [14] applied standard image processing techniques (e.g., enhancement, filtering) before classification. This preprocessing step significantly improved the performance of their ML model in identifying DR vs. non-DR images

B. Dataset-Based Comparative Studies :

X. Li et al. [15] provided a comparative evaluation of DR detection models using datasets such as Messidor, EyePACS, and IDRiD. Their work emphasized the importance of standardized evaluation metrics and the impact of dataset imbalance.

H. Jiang et al. [16] reviewed the properties and challenges of using fundus image datasets for DR detection. The review highlighted how dataset quality directly affects model generalizability and clinical applicability.

C. Review and Meta Analysis Studies :

W. Zhang et al. [17] and K. Xu et al. [18] conducted broad surveys on DR detection techniques, classifying approaches into binary, multi-stage, lesion-based, and vessel-based strategies. A. Mane et al. [19] emphasized the need for scalable, automated DR screening systems. Their review suggests leveraging AI to triage patients efficiently, reducing clinical workload and enhancing accessibility in under-resourced areas.

Figure 1 shows Diabetic Retinopathy detection integrates federated learning, GAN-based data augmentation, and CNN-based classification to enhance the accuracy and robustness of retinal image analysis. The complete pipeline is designed to address the challenges of data scarcity, privacy preservation, and heterogeneous data distribution across medical centers.

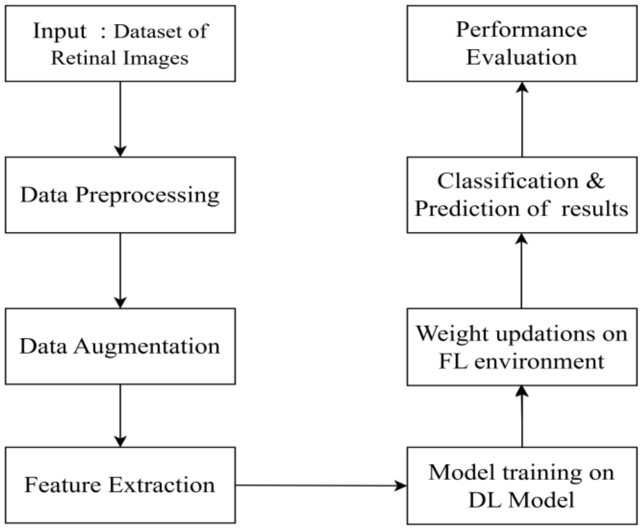

Fig. 1. System Architecture

1) Input and Preprocessing :

The system accepts retinal scan images as input, which may be collected from multiple sources. These images are then forwarded to main processes:Federated Learning, Augmentation and Classification .

2) DR Classification Legend:

Classification of DR severity, the architecture incorporates a five-class system as follows:No DR, Mild DR, Moderate DR,Severe DR,Proliferative DR etc.This classification provides a granular framework for the detection and grading of DR based on image features.

3) Federated Learning-Based Model Update:

In the federated learning branch, input images are distributed across multiple clients: Kaggle Client, Eyepacs Client, and Messidor Client. Each client performs local updates on its data using a shared CNN model.The local model weights are then aggregated at a central Federated Server, ensuring privacy-preserving model training without transferring raw patient data.

The updated global model weights are sent back to each client in an iterative cycle, enhancing model generalization across different data domains.

4) GAN-Based Data Augmentation :

Simultaneously, the input image is processed through a Generative Adversarial Network for data augmentation. This module generates synthetic yet realistic retinal images to address class imbalance and improve the diversity of training samples.The augmented dataset enhances the training of the CNN classifier by providing a richer variety of pathological features, particularly in underrepresented classes.

5) CNN-Based Classification :

The CNN model, trained using both original and GAN-augmented images, performs the classification of DR presence and severity. The final layer of the CNN outputs a class prediction that falls into one of the five predefined DR categories.

6) Result Classification :

The final stage involves interpreting the model output into multiple classes.This result is useful for both automated screening tools and decision-support systems in clinical settings.

IV. PROPOSED METHODOLOGY

Below mentioned algorithm for Diabetic Retinopathy Detection with CPU vs GPU Handling is a decentralized learning approach where several clients with their own private retinal image datasets collectively train a global convolutional neural network (CNN) without exchanging their raw data. In each of the T rounds of communication, a subset of clients is chosen to do local training. Each client downloads the latest global model weights and decides whether to train on a GPU or CPU as per device availability. They then train the model locally for E epochs with mini-batches of DR images, conducting forward propagation, loss calculation (e.g., cross-entropy), backpropagation, and weight updates. The updated weights are sent back to the server after local training. The server then computes these weights in a weighted average form depending on the data size of each client to update the global model.This is repeated for rounds until the last round, leaving us with a properly trained CNN model that can diagnose diabetic retinopathy from retinal images without compromising data privacy among clients.Output of algorithm is classification of diabetic retinopathy in five classes. Due to the use of federated learning multiple clients are involved. Main contribution of federated learning is to provide privacy to data.As model is going towards clients rather than data transferring to training models.

V. RESULT ANALYSIS

Below Figure 2 and 3 show comparison between a CNN and an existing algorithm in terms of Accuracy and Precision values. From the Accuracy graph (left), it is apparent that a CNN has an apparent lead, having near 97% accuracy over the performance of an existing algorithm at just over 91%, which means that the CNN makes correct predictions much more regularly. In the Precision chart (right), the CNN also surpasses an existing algorithm, almost reaching 100% precision, whereas an existing algorithm remains considerably lower, around 80% only.

Algorithm : Federated CNN Training for DR Detection with CPU vs GPU Handling

Input: T: Number of communication rounds,
E: Number of local epochs,
η: Learning rate
W_0: Initial global CNN model weights,
C: Set of clients with DR image datasets
Output: Final global CNN model weights W_T

Procedure: **Initialize** global CNN model weights W_0
For each communication round t = 1 to T do
Select a subset of clients S_t ⊆ C
For each client k ∈ S_t in parallel do
Client downloads current global CNN weights W_t
If client has GPU:
Assign device ← GPU
Else
Assign device ← CPU
Initialize local CNN model: W_k ← W_t
For each local epoch e = 1 to E do
For each mini-batch b of retinal images and labels in local data do
Forward propagate batch b through CNN on device
Compute loss (e.g., cross-entropy for classification)
Backpropagate to calculate gradients ∇
Update weights: W_k ← W_k - η * ∇
End For
End For
Client sends updated CNN weights W_k to server
End For
Server aggregates updates using weighted average:
W_{t+1} ← ∑_{k ∈ S_t} (n_k / n_total) * W_k (where n_k is the number of DR images on client k)
End For
Return final global CNN model weights W_T

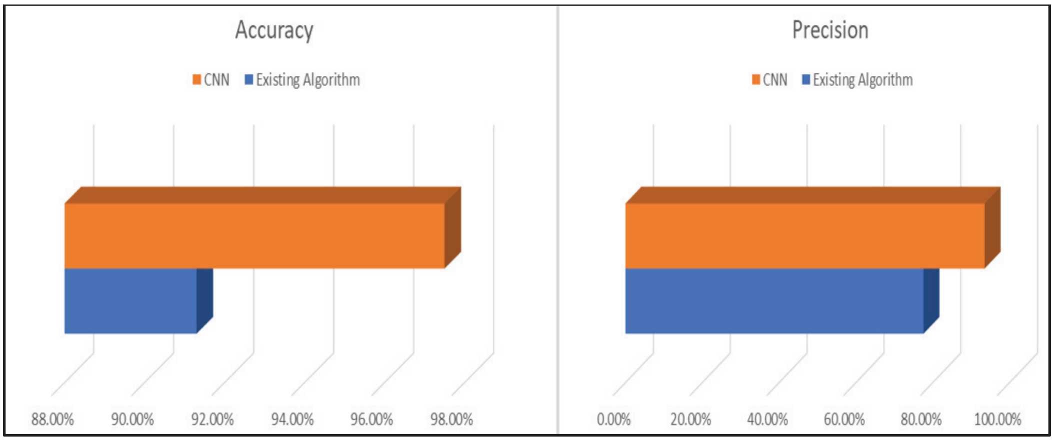

Fig. 2. Bar representation of Accuracy , Fig 3. Bar representation of Precision

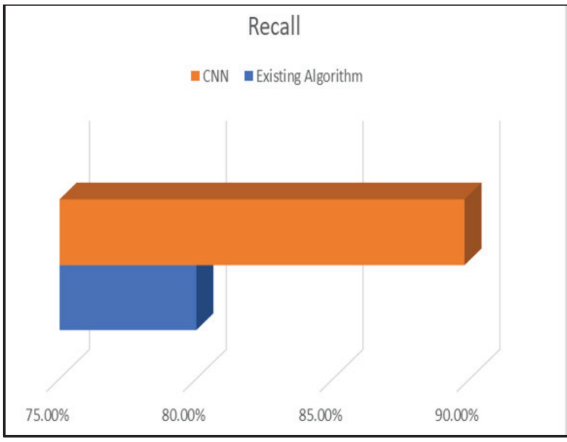

Fig 4. Bar representation of Recall ,

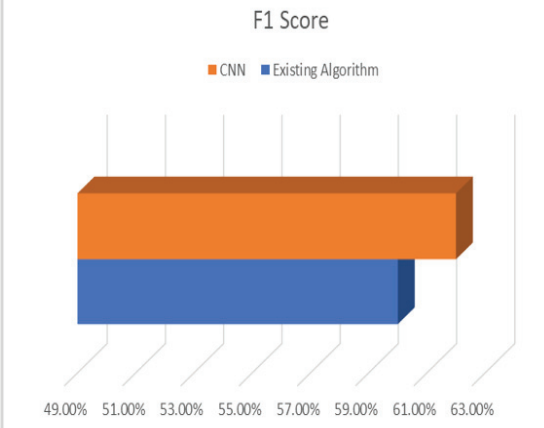

Fig 5. Bar representation of F1 score

Below Figure 4 and 5 shows the performance of a Convolutional Neural Network with an existing method in terms of Recall and F1 Score measures. In the Recall plot (left), the CNN outperforms the existing method by a significant margin, having a nearly 90% recall rate compared to the existing method's approximately 80% recall rate, which shows that the CNN is more accurate in identifying positive cases. Likewise, in the F1 Score chart (right), the CNN once more exhibits better performance, almost at 63% whereas the existing algorithm hovers at around 59%.

VI. CONCLUSION

This paper presents a comprehensive review of recent advancements in automated Diabetic Retinopathy (DR) detection, followed by a proposed system architecture that leverages emerging technologies to address key challenges in medical image analysis. The proposed framework integrates federated learning for privacy-preserving model training, Generative Adversarial Networks (GANs) for class-balanced data augmentation, and Convolutional Neural Networks (CNNs) for robust DR classification.By incorporating federated learning, the architecture enables collaborative training across decentralized data sources such as Kaggle, Eyepacs, and Messidor datasets, without compromising patient privacy. The GAN module addresses data imbalance by generating synthetic retinal images, thus enriching the diversity of training samples. CNN-based classification, trained on both real and augmented data, ensures accurate and granular detection across five standardized DR severity levels.Overall, the architecture not only enhances diagnostic precision but also aligns with the future of scalable, privacy-aware, and data-efficient clinical decision support systems for DR screening. Future work may explore real-time deployment, uncertainty estimation, and integration with mobile health platforms.

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